EXPERIMENT – 2

Encryption – Crypto 101 using TryHackMe Platform

Aim

To understand the fundamentals of **cryptography** by performing hands-on exercises in the **Crypto 101** room on the **TryHackMe** platform, including basic encryption, encoding, hashing, and cryptographic concepts.

Tools / Platform Required

- Web browser (Chrome / Firefox)
- Internet connection
- TryHackMe account (free)
- Built-in TryHackMe terminal (AttackBox) or local Linux machine

Theory (Brief)

Cryptography is the practice of securing information by converting plain text into unreadable format (cipher text) using encryption techniques.
Crypto 101 introduces basic concepts such as:

- Encoding vs Encryption
- Symmetric and Asymmetric encryption
- Hashing
- Common ciphers (Caesar, Base64, etc.)

Steps to Perform the Experiment

Step 1: Login to TryHackMe

1. Open browser and go to: https://tryhackme.com
2. Login using your registered credentials.

Step 2: Open Crypto 101 Room

1. In the search bar, type **Crypto 101**
2. Click on the room named **Crypto 101**
3. Click **Join Room**

Step 3: Start the Machine (if required)

1. Click **Start Machine / Start AttackBox**
2. Wait for the virtual environment to load.

Step 4: Perform Cryptography Tasks

Complete the tasks provided in the room, such as:

4.1 Encoding & Decoding

(a) Base64 Encoding / Decoding

Purpose:
Base64 is an **encoding technique** used to represent binary data in readable ASCII format.

How to Solve

To Decode Base64

1. Copy the given Base64 string from the question.

2. Open the TryHackMe terminal.

3. Use the command:

echo "Q3J5cHRvZ3JhcGh5" | base64 -d

4. The output will be the decoded plain text.

5. Enter the decoded text into the answer box.

To Encode Base64

echo "cryptography" | base64

Expected Output Example: Cryptography

(b) Hexadecimal Conversion

Purpose:
Hexadecimal is a base-16 encoding system commonly used in computing**.**

How to Solve

Hex to Text

echo 48656c6c6f | xxd -r -p

Text to Hex

echo "Hello" | xxd -p

Expected Output: Hello

4.2 Classical Ciphers

(a) Caesar Cipher

Purpose:
Caesar cipher shifts letters by a fixed number.

How to Solve

1. Identify the shift value (given or by trial).

2. Use online Caesar decoder or terminal tool.

Using terminal
echo "KHOOR" | tr 'A-Z' 'X-ZA-W'

(Shift = 3)
Expected Output: HELLO

(b) Shift-Based Encryption

Purpose:
Similar to Caesar cipher but may use different shift values.

How to Solve

1. Try different shift values until meaningful text appears.

2. Online tools can be used if allowed.

Example:

- Shift 1 → incorrect
- Shift 3 → readable text → correct answer

4.3 Hashing

(a) Understanding Hash Algorithms

Hash Type	**Length**
MD5	32 characters
SHA-1	40 characters
SHA-256	64 characters

(b) Identifying Hashes

How to Solve

1. Count the number of characters in the given hash.
2. Match it with the table above.

Example:

5d41402abc4b2a76b9719d911017c592

→ 32 characters → MD5

Generating Hashes

```
echo -n "hello" | md5sum
echo -n "hello" | sha1sum
echo -n "hello" | sha256sum
```

4.4 Encryption Concepts

(a) Encryption vs Hashing

Encryption	**Hashing**
Reversible	Irreversible

Uses keys	No keys
Used for secure communication	Used for password storage

(b) Symmetric vs Asymmetric Encryption

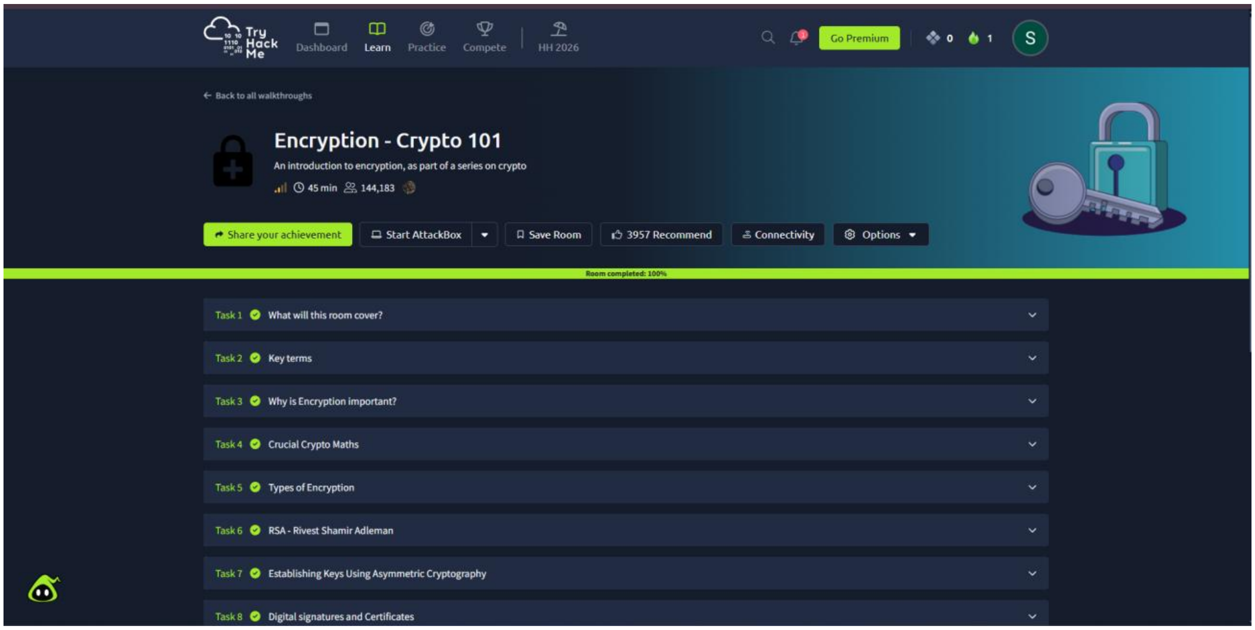
Symmetric	**Asymmetric**
One key	Two keys
Fast	Slower
Example: AES	Example: RSA

Step 5: Verify Completion

1. Ensure all tasks are marked **Completed**
2. Observe the success message at the end of the room

Output

- Successfully decoded encrypted messages
- Paste the screen shots

- 

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Result

Thus, the **Crypto 101** experiment was successfully completed on the **TryHackMe platform**.
The experiment helped in understanding **basic cryptographic concepts**, encryption techniques, encoding methods, and hashing algorithms through hands-on practice.